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RESEARCH ARTICLE

Cross-Subject EEG-Based Emotion Recognition Using Deep Metric Learning and Adversarial Training

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ABSTRACT Nowadays, due to individual differences and the non-stationarity properties of EEG signals, developing an accurate cross-subject EEG emotion recognition method is in demand. Despite many successful attempts, the accuracy of generalized models across subjects is inferior compared to those limited to a specific individual. Moreover, most cross-subject training methods assume that the unlabeled data from target subjects is available. However, this assumption does not hold in practice. To address these issues, this paper presents a novel deep similarity learning loss specific to the emotion recognition task. This loss function minimizes intra-emotion class variations of EEG segments *with different subject labels* while maximizing inter-emotion class variations. Another key aspect of the proposed semantic embedding loss is that it preserves the order of emotion classes in the learned embedding. Specifically, it ensures that the embedding space maintains the semantic order of emotions. Also, we integrate the deep similarity learning module with adversarial learning, which helps to learn a subject-invariant representation of EEG signals in an end-to-end training paradigm. We conduct several experiments on three widely used datasets: SEED, SEED-GER, and DEAP. The results confirm that the proposed method effectively learns a subject invariant representation from EEG signals and consistently outperforms the *state-of-the-art* (SOTA) peer methods.

INDEX TERMS EEG signals, cross-subject emotion recognition, deep metric learning, adversarial learning.

I. INTRODUCTION

Electroencephalography (EEG)-based emotion recognition has received immense attention in recent years. EEG signals provide a direct measurement of human emotions with excellent temporal resolution. Unlike alternative approaches such as facial expressions [1], EEG signals cannot be easily altered or deliberately restrained, thus providing reliable information.

EEG offers several advantages over other neuroimaging techniques like *functional magnetic resonance imaging* (fMRI) and *magnetoencephalography* (MEG) including [2]: 1) Portability: EEG systems are compact and portable, allowing for easy setup and data acquisition in various environments. 2) Cost-effectiveness: EEG equipment is generally more affordable compared to fMRI and MEG systems, making it more accessible to researchers, clinicians, and other professionals.

The non-stationarity of EEG signals between individual sessions and subjects poses challenges in developing practical models for EEG-based emotion recognition. This problem can be categorized into two scenarios: cross-subject and cross-session. In FIGURE 1, the inter-subject variability of emotional EEG data is depicted, highlighting the challenges involved in developing a practical EEG-based emotion recognition model. For example, with a typical multi-layer perceptron classifier, the cross-subject emotion recognition accuracy on the SEED dataset [3] can be as low as 58%. However, when the classifier is used for recognizing emotions within the same subject (i.e., intra-subject emotion recognition), the accuracy can be much higher, up to 96% [2]. The substantial decrease in emotion recognition performance

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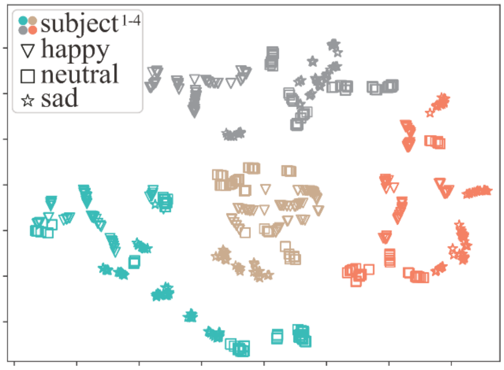


FIGURE 1. EEG signals from four different subjects, each experiencing three different classes of emotions. The large variability among the EEG signals of these subjects highlights the individual differences in the EEG patterns associated with different emotional states [4].

emphasizes the need for developing accurate cross-subject emotion recognition methods.

Although there have been many attempts to create generalized models that work across different individuals, these models are generally less accurate than those that are tailored to specific individuals.

Many cross-subject affective models utilize *domain adaptation* (DA) techniques in combination with deep neural network classifiers [5], [6]. DA aims to bridge the gap between different domains (i.e., source subjects and target subjects) by aligning the feature distributions or learning domain-invariant representations. DA techniques help mitigate the effects of individual differences and variations in EEG-based emotion representations, enabling the model to effectively recognize emotions in target subjects. This typically involves training the model on a source subject with labeled data and leveraging unlabeled (or limited) labeled data from the target subjects. However, in practice, it is not always feasible to have access to target subjects during the training stage. This poses a challenge for DA techniques that rely on target domain data for learning domain-invariant representations.

Instead of adapting the model to a specific target subject, *domain generalization* (DG) methods focus on finding domain-invariant representations from the source subjects. The goal is to learn features that capture common patterns and characteristics across different subjects in the training dataset, allowing the model to generalize well to unseen subjects without requiring access to testing subjects' signals.

Existing domain generalization (DG) methods have demonstrated acceptable accuracy on evaluated datasets. However, they suffer from the scalability aspect. These methods often necessitate constructing a separate model for each subject in the training dataset, leading to substantial time and space requirements during the training stages. Thus, developing subject-specific models poses practical challenges, especially when dealing with a large number of subjects.

To address these challenges, this paper introduces a novel *deep metric learning* (DML)-based module specifically designed for the task of emotion recognition. Our work is inspired by the successful application of DML techniques in the zero-shot learning and few-shot learning tasks, in which DML methods have shown promising results in learning discriminative representations that generalize well to unseen or novel categories with limited labeled examples.

The proposed method called, *EEG-DML*, focuses on minimizing the variations within each emotion class of EEG segments with different subject labels, while simultaneously maximizing the variations between different emotion classes. To achieve this, EEG-DML consists of a DML and an adversarial subject discriminator, learned in an end-to-end training paradigm. Here, the proposed similarity-based loss function (inside the DML module) enhances the discriminative power of the learned representation by reducing intra-emotion class variations, resulting in a compact representation within each emotion category that includes different subjects. Moreover, the loss preserves the semantic order of emotion classes in the learned embedding. By retaining this order, our method ensures that the model's predictions and representations align with the natural intensity of emotions, allowing for more accurate and smoother prediction in the emotion recognition task. On the other hand, the adversarial learning component further enhances the subject-invariant nature of the learned representation by implicitly eliminating subject-specific features.

In summary, the main contributions of the proposed method for the cross-subject EEG-based emotion recognition problem are as follows:

1- We developed a novel cross-subject emotion recognition method called EEG-DML by leveraging deep metric learning, adversarial training, and the preservation of the semantic order of emotions to improve the accuracy and generalization performance of emotion recognition models across different individuals.

2- A new similarity loss function was designed specifically for the emotion recognition task, which minimizes intra-emotion class variations of EEG segments with different subject labels while maximizing inter-emotion class variations.

3- Extensive experiments were conducted on two well-known emotion recognition datasets, namely SEED [3], SEED-GER [3], and DEAP [7] in the cross-subject setting to evaluate the performance of the proposed method. The results demonstrate the superiority of the proposed method over SOTA methods in terms of various classification metrics. Specifically, we achieved an accuracy of 91.70% on the SEED dataset, surpassing the state-of-the-art *self-organized graph neural network* (SOGNN) [8] method with an accuracy of 86.37%.

II. RELATED WORK

DA is widely used for cross-subject emotion recognition. Traditional DA methods, such as Maximum Independence Domain Adaptation (MIDA) [9], Subspace alignment (SA) [10], and Information theoretical learning (ITL) [11], were utilized in [12] for training a cross-subject machine learning model. The results obtained on the DEAP [7] and SEED datasets revealed that these techniques boosted the accuracy of the classifier by 7.25% and 13.40%, respectively, on these datasets.

Recent work utilized deep learning DA techniques and deep neural network classifiers for the cross-subject or cross-session emotion recognition tasks. Reference [13] combined autoencoders and subspace alignment to align the source domain with the target domain, thereby reducing the discrepancy between them. Similarly, the Bi-hemispheres domain-adversarial neural network (BiDANN) [14] learned two domain classifiers for each hemisphere and a global one adversarially working with an emotional classifier to learn an effective recognition model across subjects. The *regularized graph neural network* (RGNN) [15] utilized a graph neural network to learn spatial relations between different channels in EEG signals. Moreover, RGNN used node-wise domain adversarial training (NodeDAT) to handle EEG variations among subjects. Considering the differences between network layers, [16] utilized adversarial training to adapt the marginal distributions in the early layers and employed association reinforcement learning to adapt the conditional distributions in the final layers. EEGMatch [17] combined both prototype-wise and instance-wise pairwise learning. Here, prototype-wise pairwise learning focuses on capturing the global relationship between EEG data and the prototypical representation of each emotion category. In contrast, instance-wise pairwise learning considers the local intrinsic relationship among EEG signals. It captures the subtle differences and similarities within the data instances themselves, enabling a more fine-grained understanding of the underlying patterns. *Cross-Subject Source Domain Selection*(CSDS) [18] aims to select the appropriate data in the source domain for transfer learning, thereby accelerating cross-domain emotion recognition methods.

DA approaches typically require access to target domain examples, but in practice, this is not always feasible during the training stage.

Recently, domain generalization (DG) methods have gained popularity in the field of cross-subject emotion recognition. These methods are designed to reduce the dependence on data from testing subjects. DG models extract representations that are invariant across training subjects, allowing them to be applied to new subjects without requiring access to their signals.

Ma et al. [6] proposed a domain residual network for cross-subject emotion recognition. The network was designed to learn both domain-shared weights and domain-specific weights using domain-adversarial training. By capturing common patterns and shared representations across multiple subjects, the domain residual network obtained better generalization to unseen subjects. Similarly, *multi-source marginal distribution adaptation* (MS-MDA) [19] considered both domain-invariant and domain-specific features. MS-MDA constructed independent branches for each source domain of EEG data. This one-to-one domain adaptation process allowed MS-MDA to extract features specific to each source domain.

Zhao et al. [4] divided an EEG representation into two components: a private component specific to each subject and a shared emotional component that is common across all subjects. During the training phase, a shared encoder and private encoders were used to separately capture subject-invariant emotional representations and private components of the source subjects. Additionally, an emotion classifier was built on the shared partition, while individual classifiers were constructed on the combination of both partitions. In the calibration phase, this method requires only a few unlabeled EEG data from incoming target subjects to model the private components. *Denoising Mixed Mutual Reconstruction* (DMMR) [20] encoded the features of a given subject and aims to generate features similar to those of other subjects within the same emotion. *Multi-task adversarial domain adaption* (MADA) [21] found the most similar individual in the dataset to serve as the source domain. It then utilized adversarial domain adaptation to alleviate individual differences. Furthermore, MADA used multi-task learning to simultaneously classify multiple emotion dimensions, thereby capturing their inherent relations.

DG approaches face scalability issues as they require building a specific model for each subject in the source domain, which results in significant time and space requirements during the training stages.

Some approaches utilized contrastive learning [22] for the cross-subject EEG emotion recognition task. Mohsenvand et al. [23] proposed a method inspired by the SimCLR framework [22] that enforced the model to learn similarities between different views of augmented EEG signals. To augment the samples, various techniques were employed, such as temporal masking, linear scaling, and adding Gaussian noise. The pre-trained model, trained on a combination of three datasets namely SleepEDF, TUH, and SEED, demonstrated good performance on multiple downstream tasks, including sleep stage classification, clinical abnormality detection, and emotion recognition.

Banville et al. [24] utilized temporal context prediction and contrastive predictive coding techniques for learning EEG representations on two clinical datasets. By leveraging the contrastive learning strategy and utilizing large datasets during the pretraining phase, the pre-trained model attained the ability to reduce the impact of noise or irrelevant information present in the EEG signals.

Shen et al. [2] employed contrastive learning in the pre-training stage to maximize the similarity of EEG segment representations when subjects experienced the same emotional stimuli, as opposed to different stimuli.

Similarly, our method proposes a novel semantic embedding loss that enforces the feature extractor to embed the input signal into a cross-subject emotion space. However, it jointly works with the classification loss and retains the order of emotions classes in the learned embedding (*negative* < *neutral* < *positive*). By capturing this order, the model can represent the underlying emotional variations more effectively.

III. MATERIALS AND THE PROPOSED METHOD

A. PROBLEM STATEMENTS

To simplify the notation, we define $[n] = \{1, 2, \ldots, n\}$ for an arbitrary natural number n. In the cross-subject emotion recognition, the subjects in training and test datasets are different. More precisely, we assume a training dataset $D_{train} = \{(\boldsymbol{x}_i, y_i, l_i)\}_{i\in[N]}$ is available where $\boldsymbol{x}_i$ is the input EEG signal, y_i indicates the emotion label and $l_i \in L_{train}$ shows the subject label. In the test phase, we evaluate the model on $D_{test} = \{(\boldsymbol{x}_i, y_i, l_i)\}_{i\in[N_{test}]}$ where $l_i \in L_{test}$. In this setting, $L_{train} \cap L_{test} = \phi$.

The proposed method utilizes both deep metric learning and adversarial techniques to learn a subject-invariant representation for EEG signal segments. FIGURE 2 illustrates the main components of the proposed model.

The feature extraction module in the proposed architecture aims to extract discriminative cross-subject emotion information from the input EEG signal. This module is responsible for capturing both the temporal and spatial characteristics of the EEG signals, enabling the model to learn meaningful representations that can differentiate between different emotions.

As observed, our architecture involves a semantic embedding loss. The proposed *similarity-based loss function* enforces the feature extractor to embed the input signal into a cross-subject emotion space. The learned embedding minimizes intra-emotion class variations of EEG segments *with different subject labels* and maximizes inter-emotion class variations. Moreover, we have placed a projection head between the feature extractor and the loss. The projection head transforms input features into a low-dimensional embedding, enhancing the effectiveness of our similarity-based loss. Additionally, it mitigates a high correlation between the classifier and the DML module.

The emotion classifier is a Fully Connected (FC) layer followed by a Softmax. It should identify the emotion of the given EEG feature vector. The subject discriminator tries to identify the subject of the obtained feature vector. However, this objective contrasts with our goal to learn general discriminant feature representations for all subjects. To remove the subject-related features, we need to *maximize* the subject discrimination loss, which motivates a minimax game between our feature extractor and the subject discriminator. In the following, the elements of our architecture are discussed in detail.

B. FEATURE EXTRACTOR

We extract *Differential Entropy* (DE) features from the raw EEG signal as many advanced studies [2], [8], [19] have demonstrated the effectiveness of these features for the cross-subject emotion recognition task. These features represent the logarithm energy spectrum within a particular frequency band. Specifically, for a frequency band $\boldsymbol{x}$, they are calculated as:

$$DE = -\int p(x) \log(p(x))\, dx$$

These features are passed to the feature extractor module, which aims to learn meaningful representations of the input signals for discriminating between different emotions. To achieve this, we utilized the self-organized graph neural network (SOGNN) [8] as the feature extractor. SOGNN can capture relationships and dependencies among different channels using a graph structure. Moreover, the self-organized aspect of this model facilitates adapting the graph structure based on the input signals. Additionally, the network obtained SOTA results in the cross-subject emotion recognition task on the SEED dataset. We denote the deep feature extractor by the function $\boldsymbol{\phi}(.; \boldsymbol{\theta}_f)$ parameterized by $\boldsymbol{\theta}_f$.

C. DML MODULE

This module contains a novel semantic embedding loss and a projection head positioned between the feature extractor and the loss. The projection head denoted by $\boldsymbol{h}(.; \boldsymbol{\theta}_h)$ projects extracted features into a low-dimensional embedding. Since in a high-dimensional space, the Euclidean distance becomes meaningless, projecting features to a low-dimensional space promotes the quality of our distance-based loss function. Also, the projector reduces the correlation between the semantic embedding loss and the classification head. We implemented the projection head using a multilayer perceptron (MLP) with one hidden layer (FC+ReLU).

DML module presents a novel semantic embedding loss that enforces the feature extractor to embed the input signal into a cross-subject emotion space. The learned embedding minimizes intra-emotion class variations of EEG segments with different subject labels and maximizes inter-emotion class variations.

Assume our dataset has three emotion classes named negative (-1), neutral (0), and positive $(+1)$. Note that this assumption holds for some commonly used datasets such as SEED [3], SEED-GER [3], SEED-FRA [3], and DEAP [7].

Since each emotion class has different intensity levels, the proposed method considers multiple proxies per class, trained simultaneously with other network parameters using the *backpropagation* (BP) mechanism. Let $\{\boldsymbol{w}_k^c\}_{k=1}^K$ denote the K proxies of class c. We define the distance between EEG signal $\boldsymbol{h}_i = \boldsymbol{h}(\boldsymbol{\phi}(\boldsymbol{x}_i))$ in the embedding space and the emotion class c as follows:

$$d_{i,c} = \sum_{k\in[K]} s_{i,k}^c \left\| \boldsymbol{h}_i - \boldsymbol{w}_k^c \right\|_2^2 \tag{1}$$

where $s_{i,k}^c$ shows the normalized similarity of proxy $\boldsymbol{w}_k^c$ to $\boldsymbol{h}_i$ defined as:

$$s_{i,k}^c = \frac{\exp\left(\gamma \boldsymbol{h}_i^\top \boldsymbol{w}_k^c\right)}{\sum_{j\in[K]} \exp\left(\gamma \boldsymbol{h}_i^\top \boldsymbol{w}_j^c\right)}. \tag{2}$$

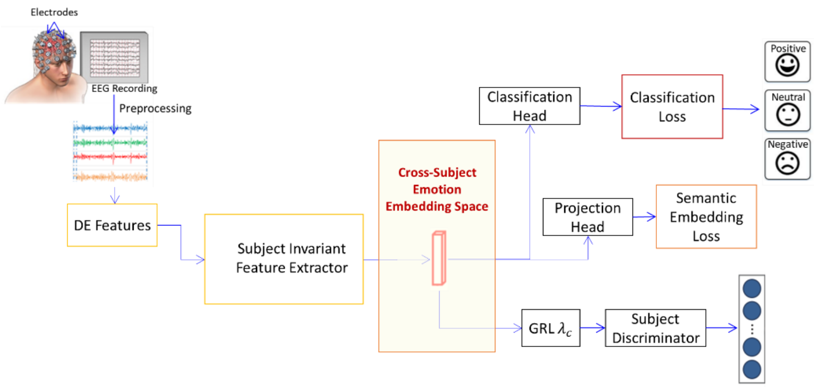


FIGURE 2. Architecture and the learning process of the proposed subject-invariant EEG-based emotion recognition method.

We aim to learn the emotion embedding such that each positive or negative EEG signal $\boldsymbol{h}_i$ with the emotion class y_i, satisfies the following conditions:

$$\begin{aligned} d_{i,y_i} + m_1 &< d_{i,0}, \\ d_{i,0} + m_2 &< d_{i,-y_i}, \end{aligned} \tag{3}$$

where m_1 and m_2 show the margin hyperparameters. These constraints push EEG signals from different subjects to their emotion classes (positive or negative) and keep them away from other classes by a margin. Also, the similarity relations between emotion classes are preserved, following the order of (*negative* $<$ *neutral* $<$ *positive*). For the neutral class, we define the constraints as follows:

$$\begin{aligned} d_{i,0} + m_1 &< d_{i,+1}, \\ d_{i,0} + m_1 &< d_{i,-1}. \end{aligned} \tag{4}$$

Satisfying these constraints ensures that the intra-distance within the neutral class (*which consists of different subjects*) is minimized, effectively keeping the neutral signals away from both positive and negative classes.

Let $\mathcal{B} = \{(\boldsymbol{x}_i, y_i) | i \in [B]\}$ show the current mini-batch sampled from D_{train}. To satisfy the constraints, we propose the following loss function:

$$\begin{aligned} \mathcal{L}_{sim}(\mathcal{B}) = \sum_{i=1}^{B} d_{i,y_i} &+ \sum_{y_i \in \{-1,+1\}} \left[d_{i,y_i} + m_1 - d_{i,0}\right]_+ \\ &+ \left[d_{i,0} + m_2 - d_{i,-y_i}\right]_+ + \sum_{y_i=0} \left[d_{i,0} + m_1 - d_{i,+1}\right]_+ \\ &+ \left[d_{i,0} + m_1 - d_{i,-1}\right]_+, \end{aligned} \tag{5}$$

where $[\cdot]_+ = \max\{., 0\}$. The first term in the loss function minimizes intra-class distances within each emotion class, irrespective of the subject label. The second term pushes each positive/negative EEG signal away from neutral and opposite emotion classes by the specified margins. The last term keeps each neutral EEG signal at a distance from positive and negative classes.

D. EMOTION CLASSIFIER

The output of the feature extractor is passed to the emotion classifier to predict the emotion label. We can implement this module using an FC layer followed by the Softmax activation. To train the classifier, we adopt the standard Cross-entropy loss function. Let $\boldsymbol{t}_i = \text{one} - \text{hot}\,(y_i + 1)$ be the true class of signal i in one-hot encoded format, and t_{ij} represent the j-th component of $\boldsymbol{t}_i$. Then, the loss is computed as:

$$\mathcal{L}_c(\mathcal{B}) = \sum_{i=1}^{B} \sum_{j \in [C]} -t_{ij} \log\left(p_{ij}\right) \tag{6}$$

where C is the number of emotion classes and p_{ij} represents the predicted probability of signal i belonging to emotion class j.

E. SUBJECT DISCRIMINATOR

The subject discriminator aims to identify the subject from the given feature vector. We denote this module by $p_d(.; \boldsymbol{\theta}_d)$ parametrized by $\boldsymbol{\theta}_d$. This objective contrasts with learning a discriminative subject-invariant feature representation for any EEG signals. Thus, we need to maximize the discrimination loss to eliminate the subject-specific features. This idea sets up a minimax game between the feature extractor and the subject discriminator. On one hand, the feature extractor tries to deceive the discriminator by maximizing the discrimination loss, and on the other hand, the discriminator aims to discover the subject-specific information in the feature representations to identify the subject. To implement this strategy, we add a *gradient reversal layer* (GRL) between the feature extractor and the discriminator. GRL acts like an identity function during the forward stage, while during the backward stage, it multiplies the gradient by a negative coefficient $(-\lambda_d)$ and passes the results to the feature extractor.

To train this module, we choose the standard cross-entropy loss. Let $\boldsymbol{l}_i$ be the subject of signal i in the one-hot encoded format, and l_{ij} represent the j-th component of $\boldsymbol{l}_i$. Then, the loss can be formulated as:

$$\mathcal{L}_d(\mathcal{B}) = \sum_{i=1}^{B} \sum_{j \in [n_s]} -l_{ij} \log\left(p_{ij}^{(d)}\right) \tag{7}$$

where n_s is the number of subjects and $p_{ij}^{(d)}$ represents the predicted probability of signal i belonging to subject j.

F. MODEL INTEGRATION AND THE FINAL LOSS FUNCTION

In the training phase, the feature extractor cooperates with the emotion classifier to minimize the classification loss ($\mathcal{L}_d$) and the semantic embedding loss ($\mathcal{L}_{sim}$), which improves the performance of the emotion classification task in a cross-subject setting. Meanwhile, the feature extractor confuses the subject discriminator to further achieve subject-invariant representations for the given EEG signals. In contrast, the subject discriminator minimizes the discriminator loss ($\mathcal{L}_d$) to identify the true subject of the signal based on the extracted feature vector. We define the final loss in this minimax game as:

$$\mathcal{L}_f = \mathcal{L}_c + \lambda_s \mathcal{L}_{sim} - \lambda_d \mathcal{L}_d, \tag{8}$$

where the hyperparameters λ_s and λ_d balance the trade-off between the terms. We seek the saddle point in the parameter space that satisfies:

$$\begin{aligned} \boldsymbol{\theta}_f^* &= \arg\min \mathcal{L}_f\left(\boldsymbol{\theta}_f, \boldsymbol{\theta}_d^*\right), \\ \boldsymbol{\theta}_d^* &= \arg\max \mathcal{L}_f\left(\boldsymbol{\theta}_f^*, \boldsymbol{\theta}_d\right) = \arg\min \mathcal{L}_d\left(\boldsymbol{\theta}_d\right). \end{aligned} \tag{9}$$

This minimax problem can be solved using the GRL layer. FIGURE 3 illustrates the learning process of the proposed method.

G. IMPLEMENTATION DETAILS

We implemented the proposed method using the *PyTorch* deep learning library. The required preprocessing and post-processing steps are discussed in Section IV-A according to the input emotion dataset.

SOGNN contains: 1) three *Conv-Pool blocks*, 2) three *self-organized graph* layers, 3) three *graph convolution layers*, 4) one fully-connected layer, and 5) an output layer. Please refer to [8] for more details about the architecture of this network.

TABLE 1. shows the architecture of the adversarial subject discriminator module which contains a GRL layer, two FC layers each followed by a Leaky ReLU activation, and an output layer.

TABLE 1. The architecture of the subject discriminator.

Layer	Input Shape	Output Shape
GRL	$[B, d_f]$	$[B, d_f]$
FC1+Leaky Relu	$[B, d_f]$	$[B, 2d_f]$
FC1+Leaky Relu	$[B, 2d_f]$	$[B, d_f]$
Subject Classifier		
FC+ Softmax	$[B, d_f]$	$[B, n_s]$

FC: Fully Connected, B: batch size, d_f: number of features, n_s: number of subjects.

The proposed DML loss function involves two margin hyperparameters, namely m_1 and m_2. Due to the time-consuming process of adjusting these hyperparameters, we implement the proposed emotion loss using *the soft-margin triplet function* [25] that does not require no margin hyperparameter for a triplet constraint. Moreover, the experimental results in [25] showed the superiority of the soft-margin triplet function compared with several alternatives. Specifically, let $d_{xz} = d(\boldsymbol{x}, \boldsymbol{z})$ be the Euclidean distance between x and z in the learned embedding. For a given triplet $(\boldsymbol{a}, \boldsymbol{p}, \boldsymbol{n})$, the soft-margin triplet function approximates the hinge loss function $\max\left\{m - (d_{an} - d_{ap}), 0\right\}$ as:

$$s\left(d_{an}, d_{ap}\right) = \log\left(1 + \exp\left(-(d_{an} - d_{ap})\right)\right) \tag{10}$$

The architecture of the proxy-based DML module is summarized in TABLE 2.

TABLE 2. The architecture of the DML module.

Layer	Input Shape	Output Shape
Projector		
FC1+ Relu	$[B, d_f]$	$[B, 2d_e]$
FC1+ Relu	$[B, 2d_e]$	$[B, d_e]$
Proxy Layer		
FC	$[B, d_e]$	$[B, K \times n_c]$

d_e: dimension of the embedding space, n_c: number of emotion classes.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

This section deals with the experiments conducted to evaluate the effectiveness of the proposed EEG-based cross-subject emotion recognition method and compare it with other peer methods. Moreover, the ablation study of components in EEG-DML, visualization of the learned embedding, and hyperparameter analysis are presented.

We also conduct an ablation study to analyze the components of EEG-DML, a visualization experiment to illustrate the learned embedding, and a hyperparameter analysis to examine the sensitivity of the results to different settings.

A. DATASETS

1) SEED DATASET

The standardized emotional EEG dataset (SEED) is a seminal dataset in the emotion recognition field. This dataset consists of EEG recordings in 62 channels with a sampling rate of 1000 Hz from 15 healthy subjects (7 males and 8 females). FIGURE 4 illustrates the placement of electrodes on the brain surface for collecting data in the SEED dataset. The participants watched 15 movie clips each evoking a specific emotion: happy (+1), neutral (0), or sad (−1). Each clip lasted around 4 minutes. The data collection process was organized into three sessions, and each session contained all 15 trials (movie clips) for each subject.

Similar to previous studies [2], [8], [15], we adopt the pre-calculated differential entropy (DE) features smoothed by linear dynamic systems (LDS) [7] in the SEED dataset.

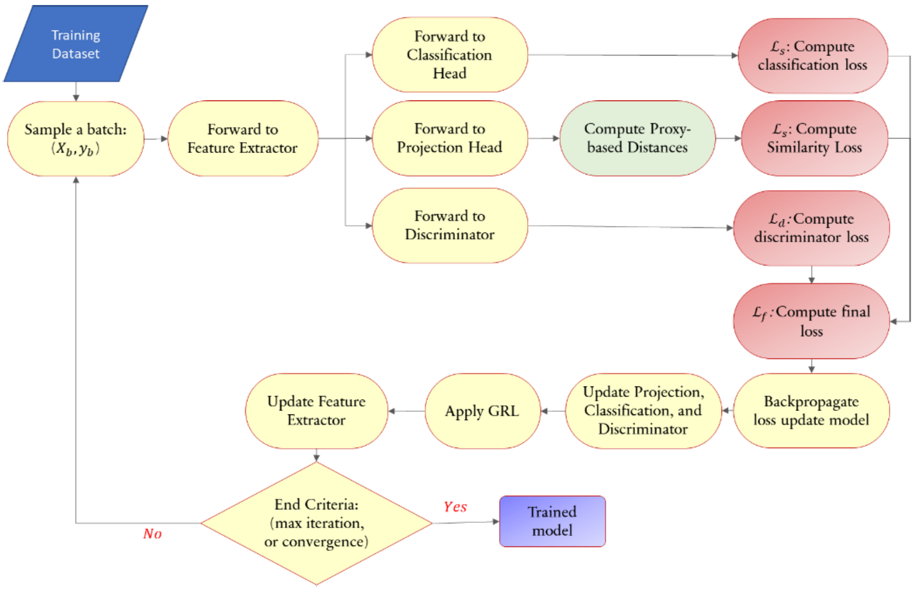


FIGURE 3. **Training flowchart of the proposed subject-invariant EEG-based emotion recognition method.**

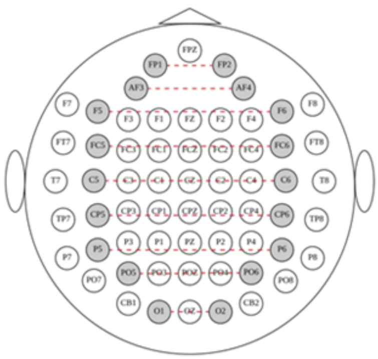


FIGURE 4. **Architecture The placement of 62 electrodes on the brain in the SEED dataset [15].**

DE features are obtained based on the Shannon entropy and quantify the complexity of a continuous random signal.

In the SEED dataset, DE features have been pre-computed across five frequency bands (delta, theta, alpha, beta, and gamma) for each second of EEG signals. These computations are independently performed for each channel and without any overlapping between consecutive signal segments. To normalize EEG signals, we compute the mean and standard deviation of all signals for each subject. Subsequently, the features of each subject are normalized by subtracting the mean and then dividing by the standard deviation.

2) SEED-GER DATASET

This dataset consists of EEG recordings in 62 channels with a sampling rate of 1000 Hz from eight German subjects (7 males and 1 female). EEG signals and eye movements were collected using the ESI NeuroScan System and SMI eye-tracking glasses. Twenty film clips (positive, neutral, and negative emotions) were chosen as stimuli used in the experiments. The duration of each film clip was around 1 to 4 minutes. Each film clip was edited to ensure emotional coherence.

Each individual took part in the experiments three times, with each session consisting of 21 trials. Before each film clip, there was a 5 sec hint, followed by watching the clip, 45 sec for self-assessment and 15 sec to rest after the clip. The order of presentation was designed in such a way that two clips with the same emotion were not shown consecutively.

In this dataset, DE features have been pre-computed across five frequency bands (delta, theta, alpha, beta, and gamma) for each second of EEG signals. These computations are independently performed for each channel and without any overlapping between consecutive signal segments. We normalized these signals in the same way as in the SEED dataset.

3) DEAP DATASET

This dataset includes EEG recordings from 32 subjects who watched 40 music video clips, each lasting one minute and designed to evoke a specific emotion. After viewing each clip, subjects rated them on a 1-9 scale for arousal, valence, liking, dominance, and familiarity. The EEG data was collected from 32 electrodes and downsampled to a frequency of 128 Hz.

We preprocessed this dataset by removing the baseline and normalizing each signal in the same way as in the SEED dataset. EEG segments with valence scores of 7 or higher were labeled as positive, those with scores between 5 and 7 were labeled as neutral, and those with scores lower than 5 are labeled as negative.

B. EVALUATION METRICS

We utilized some standard classification measures to evaluate the performance of the proposed method in the cross-subject emotion recognition task. These metrics include Accuracy, Precision, and F1-Score. Let C and n_c be the number of classes and examples per class c, respectively, and $n = \sum_{c=1}^{C} n_c$. Also, we denote *True Positive*, *True Negative*, *False Positive*, and *False Negative* of any class c by TP_c, TN_c, FP_c, and FN_c. The selected classification metrics in a multi-class problem are defined as:

$$Accuracy = \frac{\sum_{c=1}^{C} TP_c}{n}, \tag{11}$$

$$Precision_c = \frac{TP_c}{TP_c + FP_c}, \tag{12}$$

$$Recall_c = \frac{TP_c}{TP_c + FN_c} \tag{13}$$

$$(Weighted)\ Recall = \frac{\sum_{c=1}^{C} n_c\, Recall_c}{n}, \tag{14}$$

$$(Weighted)\, Precision = \frac{\sum_{c=1}^{C} n_c\, Precision_c}{n}, \tag{15}$$

$$F1 - Score = \left(\frac{2}{Recall^{-1} + Precision^{-1}}\right), \tag{16}$$

C. EXPERIMENTAL SETUP

To evaluate the performance of the proposed method in cross-subject emotion recognition, we employ the *leave-one-subject-out cross-validation* (LOSO-CV) approach. In LOSO-CV, EEG segments from one subject are left out from the training set and used as the test set, while the remaining subjects' segments are utilized for training the method. This process is repeated for each subject. Therefore, each subject's signals are used as the test set once. This approach helps to measure the performance of the competing methods on unseen subjects. We train EEG-DML using the Adam [26] optimizer, and adjust the hyperparameters of the proposed method as provided in TABLE 3.

TABLE 3. Hyperparameters of EEG-DML and their adjustment ranges.

Hyperparameter	Description	Range
n_e	maximum number of epochs	100
lr	learning rate	{1e-5, 5e-5, 1e-4}
K	number of proxies per class	{1, 3, 5, 10}
γ	inverse of Softmax temperature	{0.1, 0.5, 1, 3}
d_f	number of features	{5952}
d_e	embedding dimension	{128}
bs	*batch size*	{16}
dr	*dropout rate*	0.1
λ_s	weight of similarity loss	{0.1, 0.3, 0.5, 0.7, 1}
λ_d	weight of discriminator loss	{0.1, 0.3, 0.5, 0.7, 1}

D. RESULTS AND DISCUSSION

1) SEED DATASET

In this section, we evaluate the performance of the proposed method on the SEED dataset and compare the results with peer SOTA methods including (DMMR) [20], MADA [21](DEAP), *Cross-Subject Source Domain Selection*(CSDS) [18], *Prototypical Representation based Pairwise Learning* (PR-PL) [27], *Domain Aligned Private Encoders* (DAPE) [28], SOGNN [8], *Regularized Graph Neural Network* (RGNN) [15], and EEG-Net [29].

TABLE 4 reports the classification results of the competing methods on the SEED dataset in a Leave-one-subject-out cross-validation setting. Also, the classification results for each subject on the SEED dataset are presented in TABLE 5.

As the results indicate, EEG-DML consistently outperforms the competing methods in this dataset. We obtained an accuracy of 91.70% on the SEED dataset, surpassing the second-best method (SOGNN) with an accuracy of 86.37% (i.e., a substantial accuracy improvement of 5.33%.). The fact that our method uses the same feature extractor as SOGNN validates the effectiveness of the proposed DML module and the adversarial mechanism for cross-subject emotion recognition in this dataset. In the ablation study, we investigate the performance gain contributed by each of these components.

TABLE 4. Classification results of the competing methods in a LOSO-CV setting on the SEED dataset. The results of models reproduced by us are marked by *.

Method	Accuracy	Method	Accuracy
DMMR	88.27± 5.62	CSDS	65.80± 7.19
RP-PL	85.56± 4.35	*SOGNN	86.37 ± 4.35
DAPE	53.45±2.05	RGNN	85.30±6.72
MS-MBA	80.33±10.00	*EEG-Net	79.11±8.00
***EEG-DML**	**91.70 ± 3.19**		

2) SEED-GER DATASET

To further examine the performance of the proposed method, we evaluate it on a new emotion recognition dataset called SEED-GER in the Leave-one-subject-out cross-validation setting. The results are reported in TABLE 5. Moreover, since the confusion matrix can provide an interesting insight into the performance of a classifier, we illustrate the confusion matrices for EEG-DML and SOGNN in FIGURE 5.

3) DEAP DATASET

We also investigate the performance of EEG-DML on the DEAP dataset for valence classification in the LOSO-CV setting. We compare the results with peer SOTA methods including She et al. [30], Li et al. [31], SOGNN [8], and EEG-Net [29]. Since valence recognition in the peer methods was considered as binary classification, we convert the multi-classification results of our method to binary by considering both neutral and positive predicted labels as positive. The results are reported in TABLE 7.

The results in TABLE 6. and 7. are consistent with those of the SEED dataset and indicate EEG-DML outperforms the competing methods across all datasets. In particular, EEG-DML enhanced the accuracy of SOGNN on SEED-GER and DEAP datasets by 5.41% and 4.03%, respectively, revealing that the proposed method can learn more effective invariant representations from EEG signals in a cross-subject setting.

Additionally, EEG-DML achieved a more balanced accuracy across different emotion classes on the SEED-GER dataset. For example, SOGNN classified 73.75% of negative examples correctly and 17.5% of these instances were misclassified as positive. On the other hand, EEG-DML classified 87.50% of negative examples correctly and only 5% of them were misclassified as positive. This shows the efficacy of the proposed loss function that well discriminates different emotions from each other and captures their semantic order in the learned embedding.

We also observe that SOGNN surpassed the EEG-Net across all evaluated datasets. Thus, we can conclude that SOGNN can learn better representations in EEG-based inter-subject emotion recognition tasks compared to EEG-Net.

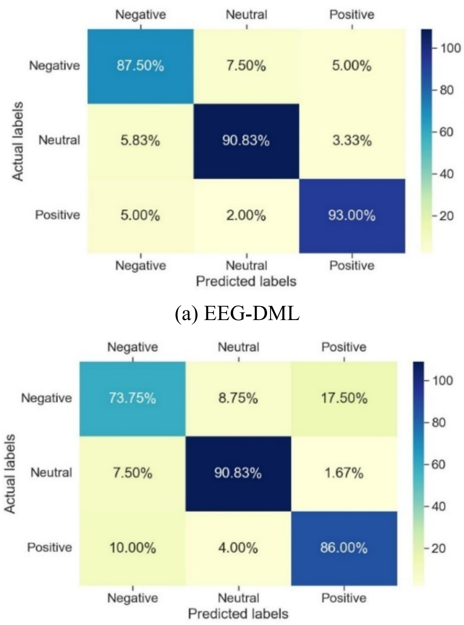


FIGURE 5. Box the mean confusion matrices of EEG-DML vs SOGNN on the SEED-GER dataset in the LOSO-CV setting.

E. ABLATION STUDY

In this experiment, we provide an ablation study of the proposed DML module and the subject adversarial component and investigate the contribution of each component individually. We employ the SOGNN as the feature extractor and derive two variants of EEG-DML as follows:

1) EEG-DML w/o DML: indicates SOGNN with the subject adversarial component. Here, we omit the DML module.
2) EEG-DML w/o Adv: shows SOGNN along with the proposed DML module. Here, we discard the subject adversarial component by setting $\lambda_d = 0$.

FIGURE 6 illustrates the results. As the results show, we can derive the following conclusions:

Both modules (i.e., Subject Adversarial and DML) improve the performance of the baseline classifier (SOGNN). For example, integrating the DML module with SOGNN increases the accuracy and F1-score of SOGNN by 5.19% and 5.34% respectively, which confirms the effectiveness of the

TABLE 5. Detailed classification results of the competing methods in a LOSO-CV setting on the SEED dataset.

	EEG-DML			SOGNN			EEG-NET		
Subj#	Acc	F1-Score	Precision	Acc	F1-Score	Precision	Acc	F1-Score	Precision
1	**91.11**	**91.23**	**91.90**	80.00	79.63	82.14	86.67	86.67	86.67
2	**86.67**	86.64	87.02	**86.67**	**86.67**	86.67	**86.67**	86.60	**87.78**
3	**93.33**	**93.14**	**94.00**	86.67	87.04	90.48	86.67	85.60	88.89
4	**91.11**	**91.18**	**91.39**	66.67	65.12	64.44	86.67	86.67	86.67
5	**91.11**	**90.99**	**91.91**	86.67	86.11	90.47	86.67	86.11	90.47
6	91.11	91.18	91.39	86.67	86.11	90.47	**93.33**	**93.27**	**94.44**
7	**93.33**	**93.27**	**94.44**	**93.33**	**93.27**	**94.44**	80.00	79.63	82.13
8	**95.56**	**95.56**	**95.56**	73.33	73.33	73.33	73.33	73.48	77.78
9	**95.56**	**95.62**	**96.08**	66.67	65.46	70.83	73.33	73.33	73.33
10	88.89	88.57	91.67	**93.33**	**93.27**	**94.44**	80.00	79.19	79.44
11	95.56	95.55	95.69	**100**	**100**	**100**	86.67	86.60	87.78
12	84.44	84.43	84.54	73.33	71.07	79.36	**93.33**	**93.27**	**94.44**
13	**91.11**	**91.11**	**91.11**	80.00	79.44	83.81	80.00	80.00	80.00
14	91.11	91.14	92.06	86.67	86.11	90.47	**100.**	**100**	**100**
15	**95.56**	**95.62**	**96.08**	93.33	93.27	94.44	86.67	86.60	87.78

TABLE 6. Classification results of the competing methods in a LOSO-CV setting on the SEED-GER dataset.

Method	Accuracy	F1-score	Precision
SOGNN	85.56 ± 2.47	85.40 ± 2.48	86.21 ± 2.81
EEG-NET	79.44±4.97	80.10 ± 4.81	82.05 ± 4
EEG-DML	**90.69 ± 1.6**	**90.77 ± 1.64**	**91.12 ± 1.48**

TABLE 7. Classification results of the competing methods in a LOSO-CV setting on the DEAP dataset (ValeNce DIMENSION). The results of models reproduced by us are marked by *.

Method	Accuracy	Method	Accuracy
She et al. [30]	65.59	Li et al. [31]	59.06
*SOGNN	63.12	*EEG-NET	60.32
***EEG-DML**	**67.15**		

proposed DML loss for learning subject-invariant representations of EEG signals. Moreover, integrating the adversarial module with SOGNN leads to an improvement in accuracy and F1-score of 3.5% and 3.5%. Thus, we can conclude that the proposed DML loss is more effective than the adversarial module in a cross-subject emotion recognition task. This finding can motivate researchers to prioritize the utilization of DML approaches rather than adversarial learning for developing a high-performance emotion recognition method for real-world scenarios.

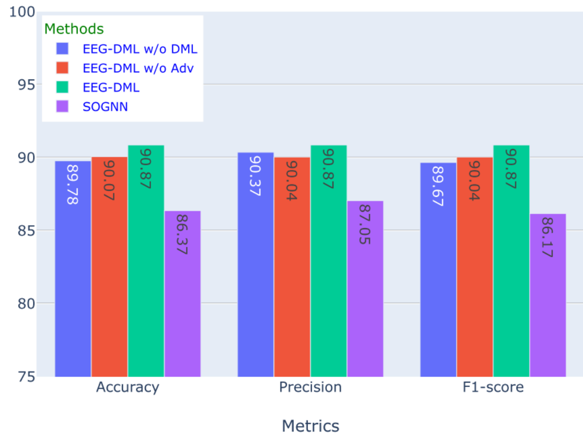


FIGURE 6. The ablation study of EEG-DML components on the SEED dataset in a cross-subject setting.

F. VISUALIZATION OF THE LEARNED EMBEDDING

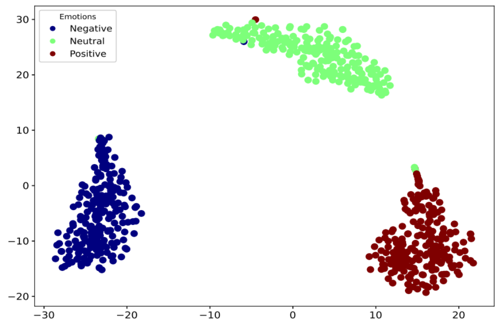
As an illustrative experiment, we visualized the SEED and SEED-German datasets in the learned embedding using t-SNE {Van der Maaten, 2008 #27}. FIGURE 7 shows these datasets in the LOSO-CV setting when the subject id is equal to 1. As the results illustrate, in the embedding space, the emotional classes in both datasets are well-discriminated and the learned embedding is subject-invariant.

G. HYPERPARAMETER ANALYSIS

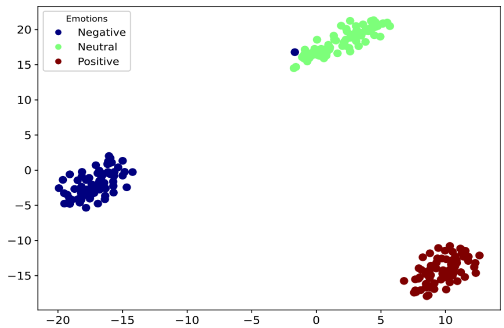
In this section, we evaluate the effects of two hyperparameters in the proposed loss function, namely K: number of proxies per emotion class and γ: that controls the sharpness of Softmax function utilized to compute the similarity of an example to an emotion class.

In the first experiment, we train EEG-DML at different values of K on the SEED dataset in the LOSO-CV setting and plot the test accuracy and F1-score for these values in FIGURE 8. Moreover, we depict the results of SOGNN in this figure to provide a better insight into the sensitivity of results to this hyperparameter.

When $K = 1$, the accuracy and F1-Score of EEG-DML are about one percent lower than those from the optimal K, which is 5 in this experiment. As the value of K increases, the performance of EEG-DML improves and peaks around K = 5. Setting K to a large value like 10 also leads to a slight decrease in the performance of the proposed method. Thus, we can conclude that by considering multiple proxies per



(a) SEED dataset



(b) SEED-GER dataset

FIGURE 7. Visualization of the SEED and SEED-GER datasets in the learned embedding datasets in the LOSO-CV setting for test subject id equal to 1.

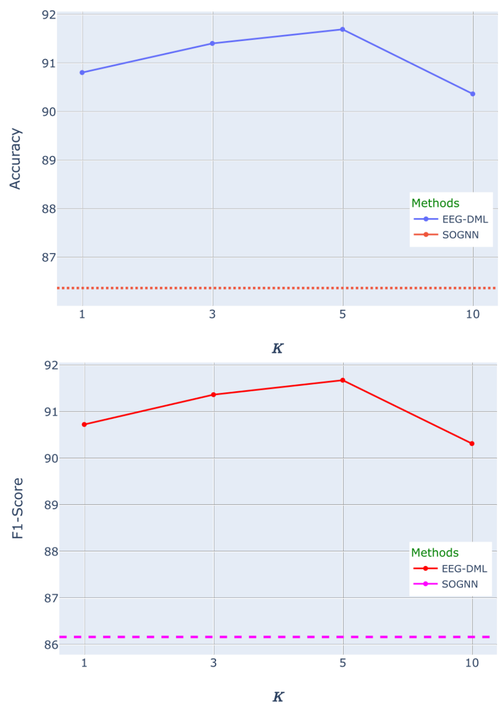


FIGURE 8. Accuracy and F1-Score of EEG-DML vs K values on the SEED dataset in the LOSO-CV setting.

emotion class in the embedding space, EEG-DML can more effectively capture the variations of EEG signals within the same emotional class. However, in a wide range of possible

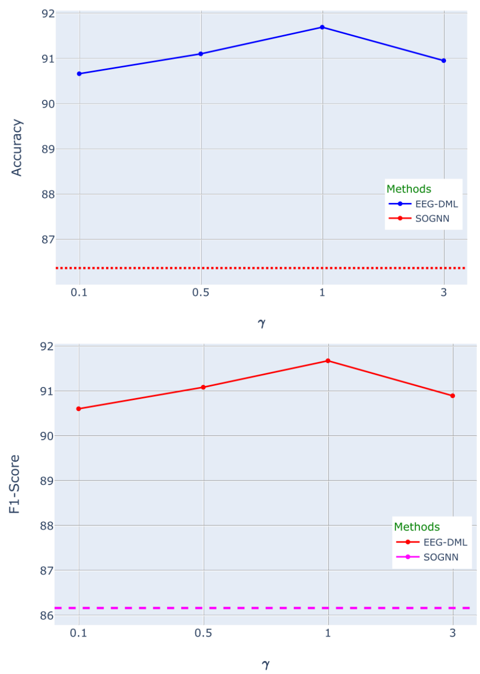


FIGURE 9. Accuracy and F1-Score of EEG-DML vs γ values on the SEED dataset in the LOSO-CV setting.

values for the K hyperparameter, EEG-DML can achieve relatively good performance, and it outperforms the SOGNN by a large margin, indicating the low sensitivity of EEG-DML to this hyperparameter.

We repeat the same experiment for the γ hyperparameter and plot the results in FIGURE 9. As the results show, the best performance of EEG-DML in this dataset is obtained by setting $\gamma = 1$. Moreover, for all evaluated values of γ, EEG-DML considerably surpasses the SOGNN, indicating the low sensitivity of the results to a specific value for this hyperparameter.

V. CONCLUSION AND FUTURE WORK

This research focuses on the task of recognizing emotions from EEG signals in a cross-subject setting. The non-stationarity of EEG signals between sessions and subjects poses challenges in developing practical applications for this problem. Here, we present a novel deep metric learning (DML)-based module specifically developed for the emotion recognition task. The proposed loss function minimizes the variations within each emotion class of EEG segments with different subject labels while increasing the variations between different classes of emotions. It also keeps the semantic order of emotion classes in the learned embedding, allowing for more accurate and smoother predictions of emotions. Additionally, we combine the DML module with an adversarial learning component to obtain a better subject-invariant representation for the EEG signals in the learned embedding by eliminating redundant subject-specific features.

We conduct several experiments on three publicly available datasets namely, SEED, SEED-GER, and DEAP in a leave-one-subject-out cross-validation setting.

The results illustrate the superiority of the proposed method called EEG-DML over SOTA methods in terms of various classification metrics. In particular, we obtained an accuracy of 91.70% on the SEED dataset, surpassing the second-best method (SOGNN) with an accuracy of 86.37%. Moreover, on the SEED-GER dataset, our method improves the accuracy of SOGNN on SEED-GER and DEAP datasets by 5.41% and 4.03%, respectively. The fact that EEG-DML uses the same feature extractor as SOGNN confirms that the proposed method can learn more effective invariant representations from EEG signals in a cross-subject setting. Additionally, by visualizing the embedding space and analyzing the confusion matrix of EEG-DML vs. SOGNN, we observe that EEG-DML obtains a more balanced accuracy across different emotion classes, validating the efficacy of the proposed loss that well discriminates different emotions from each other and captures their semantic order in the learned embedding.

However, the current work has some limitations. Particularly, it assumes that the EEG signals from unseen subjects are from the same dataset as training subjects. Moreover, it assumes the number of emotion classes is limited to three. In future work, we aim to extend our method to cross-dataset EEG-based emotion recognition tasks using a meta-learning approach. Moreover, we intend to expand our method to encompass other emotions, such as fear and anxiety. Additionally, we will study the integration of other adversarial techniques, such as [32], with the proposed similarity loss function.

DATA AVAILABILITY STATEMENT

Datasets used in the experiments are publicly available and can be downloaded from the following links: https://bcmi.sjtu.edu.cn/home/seed/ and https://www.eecs.qmul.ac.uk/mmv/datasets/deap/download.html.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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